

BUSINESS REPORT

FINANCE AND RISK ANALYTICS

PART-A

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G4 PGDSBA.O.May 23.A

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PART A

Problem Statement:

Businesses or companies can fall prey to default if they are not able to keep up their debt obligations. Defaults will lead to a lower credit rating for the company which in turn reduces its chances of getting credit in the future and may have to pay higher interest on existing debts as well as any new obligations. From an investor's point of view, he would want to invest in a company if it is capable of handling its financial obligations, can grow quickly, and is able to manage the growth scale.

A balance sheet is a financial statement of a company that provides a snapshot of what a company owns, owes, and the amount invested by the shareholders. Thus, it is an important tool that helps evaluate the performance of a business.

Data that is available includes information from the financial statement of the companies for the previous year.

Dataset for problem: [Credit Risk Dataset](#)

1.1 Outlier Treatment:

The data having 2058 rows and 58 columns has been read. There are a total 58 variables, out of which one variable i.e. 'Default' is the dependent variable. There are total 1 object variable, 4 Integer variables and 53 float variables in the data. The boxplots of the variables have been plotted to check for outliers. On checking the presence of outliers, the outliers falling in the lower range of $Q1 - (1.5 * IQR)$ and upper range of $Q3 + (1.5 * IQR)$ have been removed through Python command.

The data was also checked for duplicate values and it was found that there were no duplicates.

1.2 Missing Values Treatment:

The data was checked for missing values and it was found that there were missing values in the following variables:

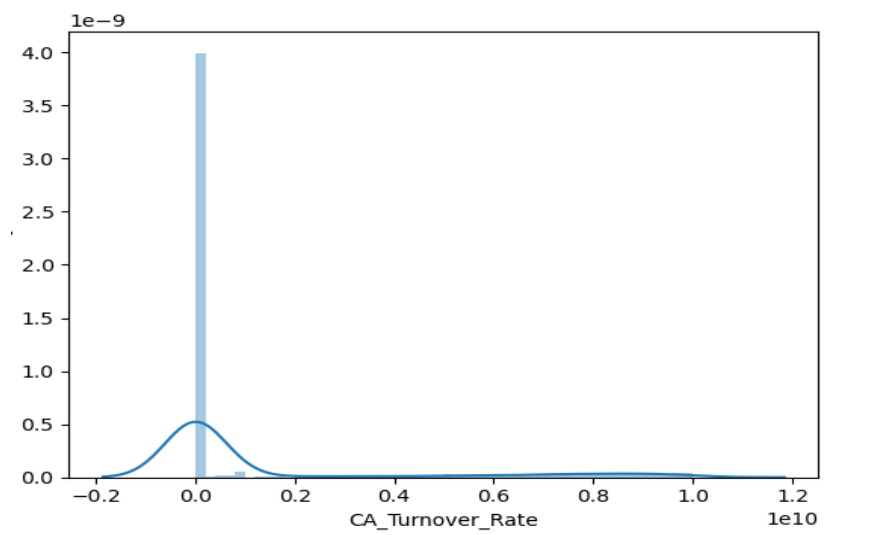
Sl. No.	Variable	Number of missing values
1	Cash_Flow_Per_Share	167
2	Total_debt_to_Total_net_worth	21
3	Cash_to_Total_Assets	96
4	Current_Liability_to_Current_Assets	14

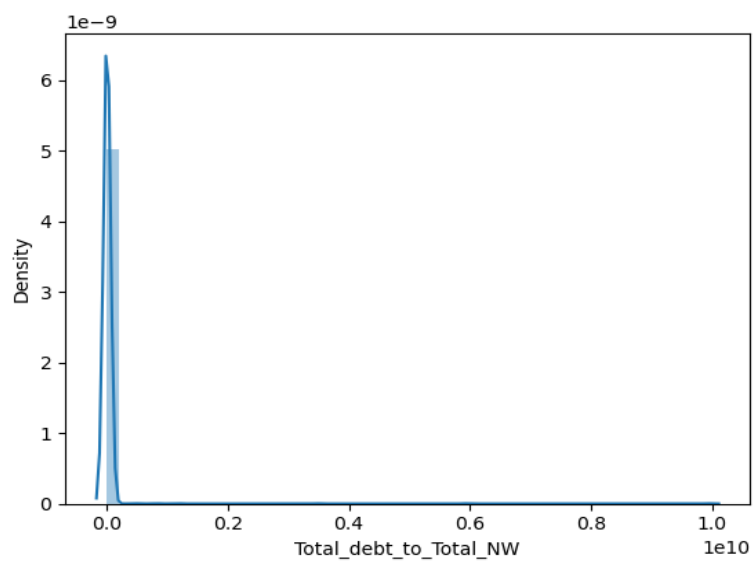
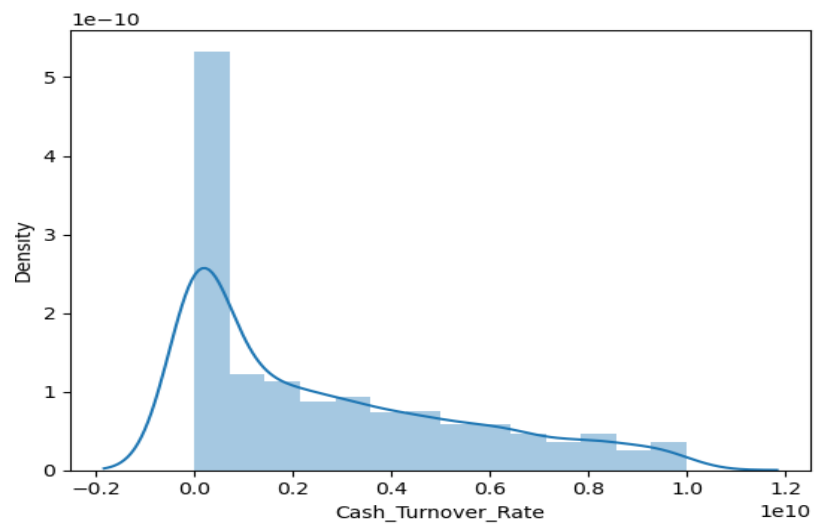
The aforesaid missing values have been deleted using Python and after deletion of missing values, there are total 1771 variables.

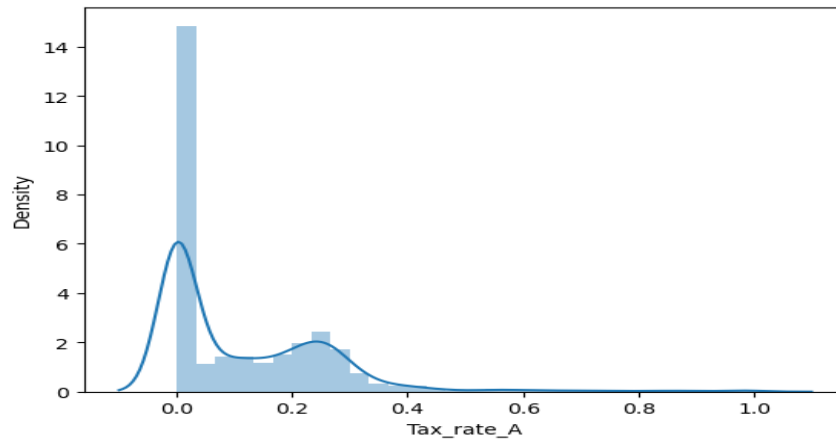
1.3 Univariate (4 marks) & Bivariate (6 marks) analysis with proper interpretation. (You may choose to include only those variables which were significant in the model building)

Univariate Analysis:

Distribution Plots (distplots) of important variables along with kde were plotted using sns library to find out the distribution of each variable. The distplots are as under:



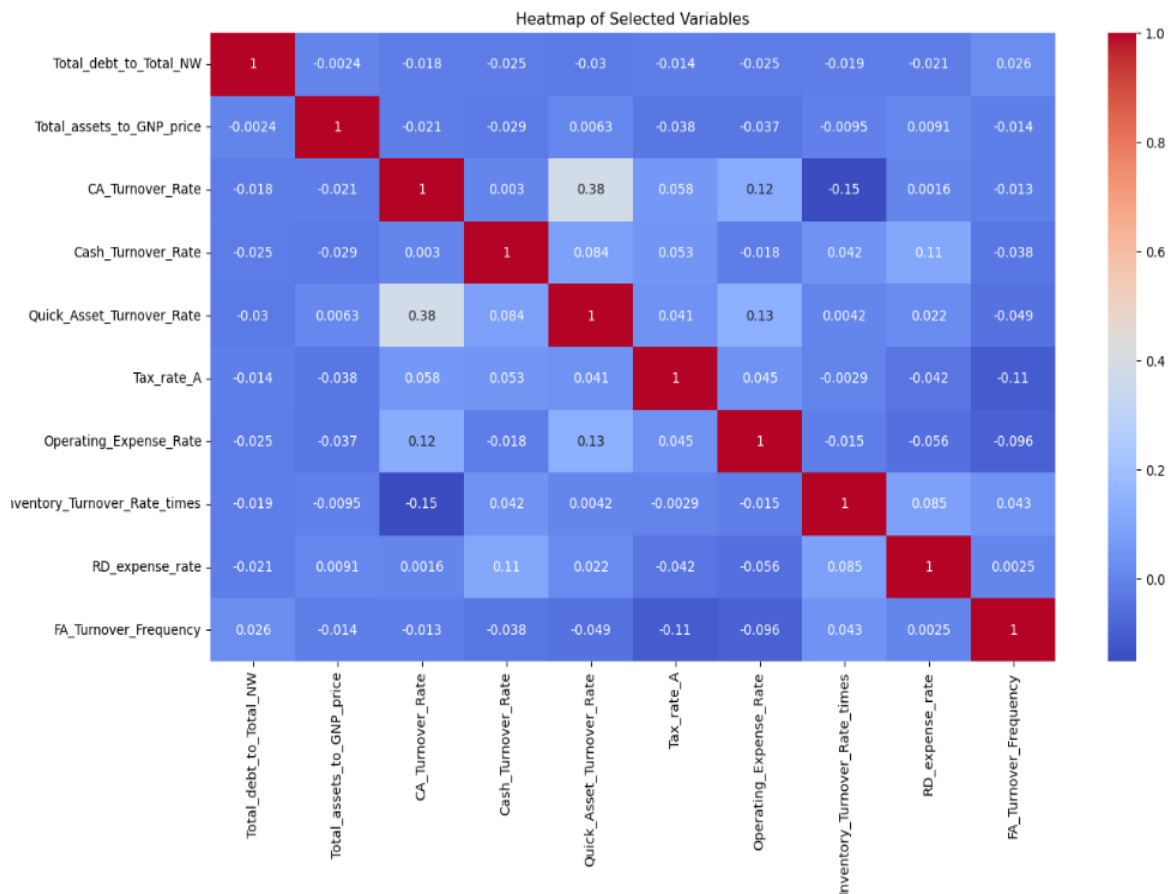


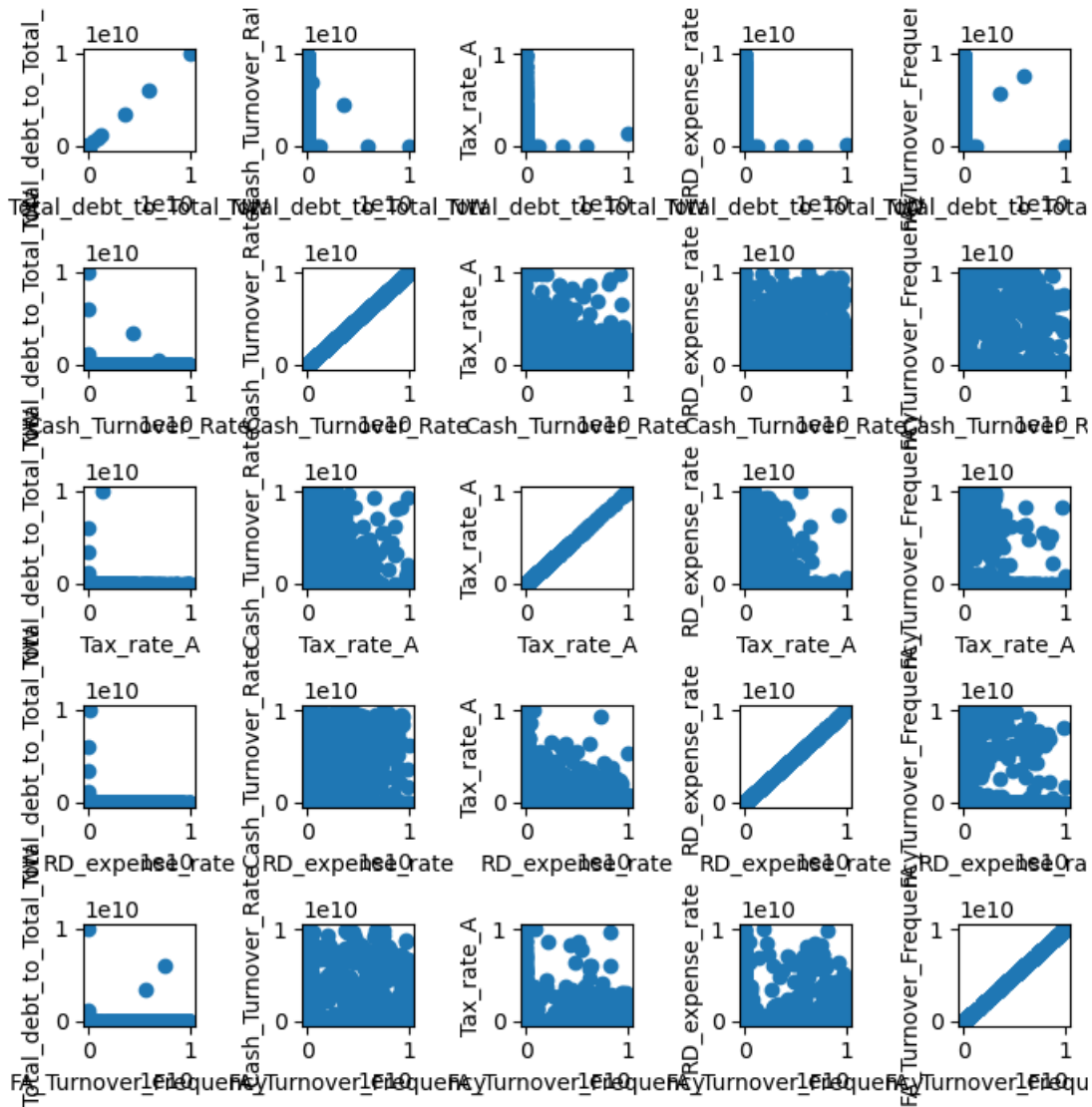


It is observed that Current Assets Turnover Rate and Total Debt to Total Net Worth follow a Normal Distribution whereas Cash Turnover Rate and Tax Rate A follow a skewed distribution with tail on the right side.

Bivariate Analysis:

Heatmap and Pairplot (scatter plot) of selected variables was drawn, which is as under:





From the Heatmap, it is seen that the variables are not correlated.

The Scatter Plots (Pairplot) also do not indicate any linear relation between the selected variables.

1.4 Train Test Split:

The variables have been checked for Variance Inflation Factor (VIF) to check for multi collinearity among variables. It was found that there are many variables having VIF values greater than 2. The variables having VIF greater than 2 at each stage have been removed.

	variables	VIF
6	Per_Share_Net_profit_before_tax_Yuan_	95.232466
19	Net_profit_before_tax_to_Paid_in_capital	94.450417
43	Cash_Flow_to_Total_Assets	43.963675
45	CFO_to_Assets	33.661770
30	Quick_Assets_to_Current_Liability	22.734497
32	Operating_Funds_to_Liability	22.152264
44	Cash_Flow_to_Liability	18.213164
2	Cash_flow_rate	17.253623
14	Current_Ratio	16.843041
25	Net_Worth_Turnover_Rate_times	16.312623
13	Cash_Reinvestment_perc	15.559450
20	Total_Asset_Turnover	15.359208
46	Cash_Flow_to_Equity	14.721958
15	Quick_Ratio	13.584544
5	Cash_Flow_Per_Share	11.324254
47	Current_Liability_to_Current_Assets	9.211593
54	Equity_to_Liability	6.479934
28	Quick_Assets_to_Total_Assets	6.188939
52	Interest_Coverage_Ratio_Interest_expense_to_EBIT	5.946950
36	Retained_Earnings_to_Total_Assets	5.383973
37	Total_income_to_Total_expense	5.373523
17	Total_debt_to_Total_net_worth	5.307666
42	Fixed_Assets_to_Assets	4.833998
16	Interest_Expense_Ratio	4.467937
31	Cash_to_Current_Liability	4.253551
29	Cash_to_Total_Assets	3.923177
8	Operating_Profit_Growth_Rate	3.805230
34	Inventory_to_Current_Liability	3.586451
9	Continuous_Net_Profit_Growth_Rate	3.548376
12	Total_Asset_Return_Growth_Rate_Ratio	3.344681
51	Degree_of_Financial_Leverage_DFL	3.328537
26	Operating_profit_per_person	3.049609
18	Long_term_fund_suitability_ratio_A	2.882527
7	Realized_Sales_Gross_Profit_Growth_Rate	2.875361
11	Net_Value_Growth_Rate	2.806192
27	Allocation_rate_per_person	2.736914
21	Accounts_Receivable_Turnover	2.736458
22	Average_Collection_Days	2.601713
38	Total_expense_to_Assets	2.325294
24	Fixed_Assets_Turnover_Frequency	2.069104

The Dependent Variable i.e. 'Default' was first taken into y data frame. Further, X and y data has been split into Training and Test data in the ratio 67:33 and Random State 42. Thereafter, X_train and y_train were concacted to form train set and X-test and y_test were concacted to form the test set for further analysis.

1.5 Build Logistic Regression Model (using statsmodels library) on most important variables on train dataset and choose the optimum cut-off. Also showcase your model building approach

Logistic Regression Model using Statsmodel library was applied to the training data having most important variables. Logit Regression Results (Model Summary- Model 1) is as under:

Model 1 Summary:

Logit Regression Results

Dep. Variable:	Default	No. Observations:	1186
Model:	Logit	Df Residuals:	1175
Method:	MLE	Df Model:	10
Date:	Thu, 28 Mar 2024	Pseudo R-squ.:	0.2358
Time:	12:38:13	Log-Likelihood:	-300.29
converged:	True	LL-Null:	-392.97
Covariance Type:	nonrobust	LLR p-value:	1.817e-34

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-3.4568	0.321	-10.753	0.000	-4.087	-2.827
Total_debt_to_Total_net_worth	137.8416	13.950	9.881	0.000	110.500	165.184
Fixed_Assets_Turnover_Frequency	12.8393	6.935	1.851	0.064	-0.754	26.432
Total_assets_to_GNP_price	23.8338	19.829	1.202	0.229	-15.030	62.697
Current_Asset_Turnover_Rate	100.1263	111.345	0.899	0.369	-118.106	318.359
Tax_rate_A	-5.6204	1.240	-4.534	0.000	-8.050	-3.191
Quick_Asset_Turnover_Rate	-6.498e-11	3.52e-11	-1.844	0.065	-1.34e-10	4.1e-12
Operating_Expense_Rate	-3.126e-11	3.44e-11	-0.908	0.364	-9.88e-11	3.62e-11
Inventory_Turnover_Rate_times	3.309e-11	3.68e-11	0.900	0.368	-3.9e-11	1.05e-10
RD_expense_rate	1.833e-10	7.66e-11	2.393	0.017	3.31e-11	3.33e-10
Cash_Turnover_Rate	-1.418e-10	4.4e-11	-3.222	0.001	-2.28e-10	-5.55e-11

After first iteration, two variables having p value>0.05 i.e. “Current Asset Turnover Rate” and “Quick asset Turnover Rate” have been dropped. After dropping the aforesaid insignificant variables, Logistic Regression was run again on Statsmodel Library. Logit Regression Results (Model Summary-Model 2) is as under:

```
: model_2.summary()
```

: Logit Regression Results

Dep. Variable:	Default	No. Observations:	1186
Model:	Logit	Df Residuals:	1177
Method:	MLE	Df Model:	8
Date:	Thu, 28 Mar 2024	Pseudo R-squ.:	0.2311
Time:	13:08:37	Log-Likelihood:	-302.17
converged:	True	LL-Null:	-392.97
Covariance Type:	nonrobust	LLR p-value:	4.762e-35

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-3.4581	0.311	-11.119	0.000	-4.068	-2.848
Total_debt_to_Total_net_worth	137.0878	13.687	10.016	0.000	110.261	163.915
Fixed_Assets_Turnover_Frequency	13.6700	6.890	1.984	0.047	0.166	27.174
Total_assets_to_GNP_price	24.0889	19.833	1.215	0.225	-14.783	62.961
Tax_rate_A	-5.6433	1.236	-4.564	0.000	-8.066	-3.220
Operating_Expense_Rate	-4.258e-11	3.4e-11	-1.252	0.210	-1.09e-10	2.41e-11
Inventory_Turnover_Rate_times	2.564e-11	3.63e-11	0.706	0.480	-4.56e-11	9.68e-11
RD_expense_rate	1.763e-10	7.61e-11	2.316	0.021	2.71e-11	3.25e-10
Cash_Turnover_Rate	-1.525e-10	4.4e-11	-3.465	0.001	-2.39e-10	-6.62e-11

After second iteration, three variables having p value>0.05 i.e. “Total Assets to GNP Price”, “Inventory_Turnover_Rate_times” and “Operating Expense Rate” have been dropped. After dropping the aforesaid insignificant variables, Logistic Regression was run again on Statsmodel Library. Logit Regression Results (Model Summary-Model 3) is as under:

```
: model_3.summary()
```

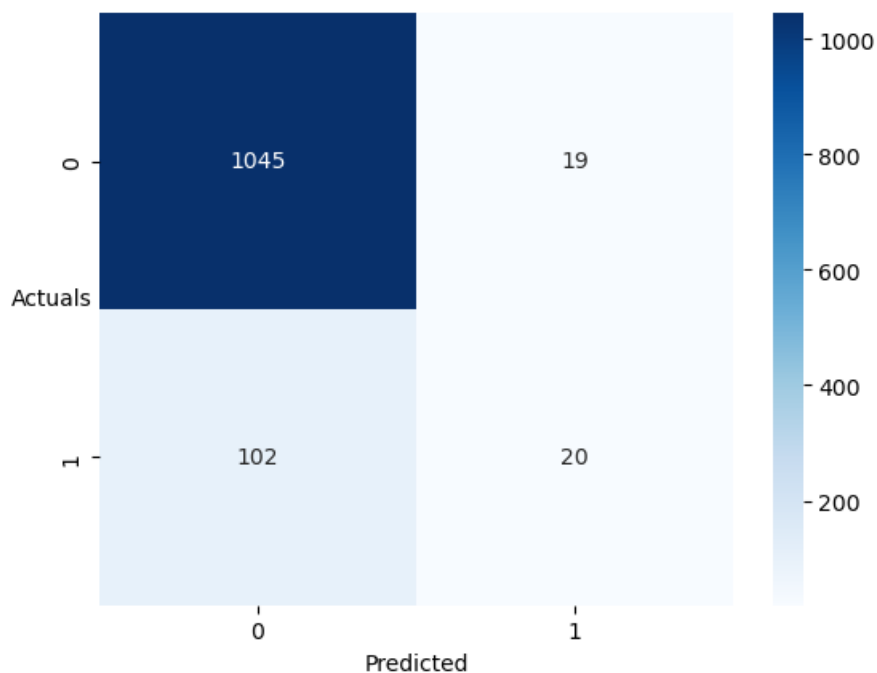
Logit Regression Results

Dep. Variable:	Default	No. Observations:	1186
Model:	Logit	Df Residuals:	1180
Method:	MLE	Df Model:	5
Date:	Thu, 28 Mar 2024	Pseudo R-squ.:	0.2270
Time:	13:08:39	Log-Likelihood:	-303.77
converged:	True	LL-Null:	-392.97
Covariance Type:	nonrobust	LLR p-value:	1.186e-36

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-3.3434	0.277	-12.064	0.000	-3.887	-2.800
Total_debt_to_Total_net_worth	135.3483	13.347	10.141	0.000	109.189	161.507
Fixed_Assets_Turnover_Frequency	15.1718	6.812	2.227	0.026	1.820	28.524
Tax_rate_A	-5.7800	1.227	-4.710	0.000	-8.185	-3.375
RD_expense_rate	1.73e-10	7.54e-11	2.296	0.022	2.53e-11	3.21e-10
Cash_Turnover_Rate	-1.514e-10	4.43e-11	-3.415	0.001	-2.38e-10	-6.45e-11

It may be seen from Model 3 Summary above that for all the variables $p < 0.05$ and all the variables are significant.

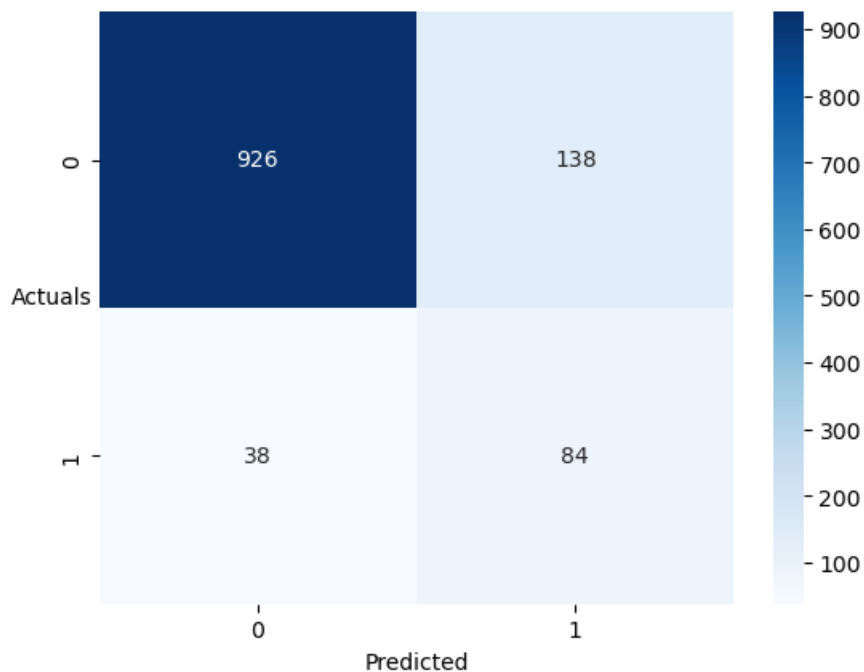
Confusion Matrix of Logit Model 3 (without optimization) has been drawn as under:



It is observed that Accuracy, Precision and Recall of the Logit Model 3 (without optimization) are 0.898, 0.513 and 0.164 respectively.

Optimal Threshold was found for the Model to be 0.155 and the Model was run on this Optimal Threshold on the Train dataset. Confusion Matrix and Classification Report for the Train Dataset are as under:

Confusion Matrix after Optimization:

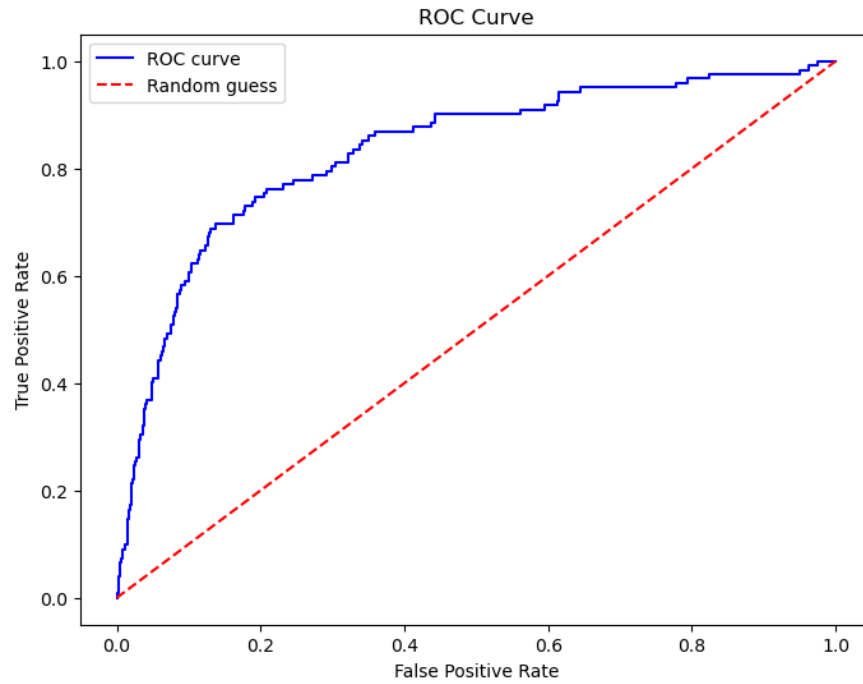


Classification Report after Optimization:

	precision	recall	f1-score	support
0	0.961	0.870	0.913	1064
1	0.378	0.689	0.488	122
accuracy			0.852	1186
macro avg	0.669	0.779	0.701	1186
weighted avg	0.901	0.852	0.870	1186

The AUC Score of the Optimized Logit Model on train dataset is 0.83 and ROC Curve is as under:

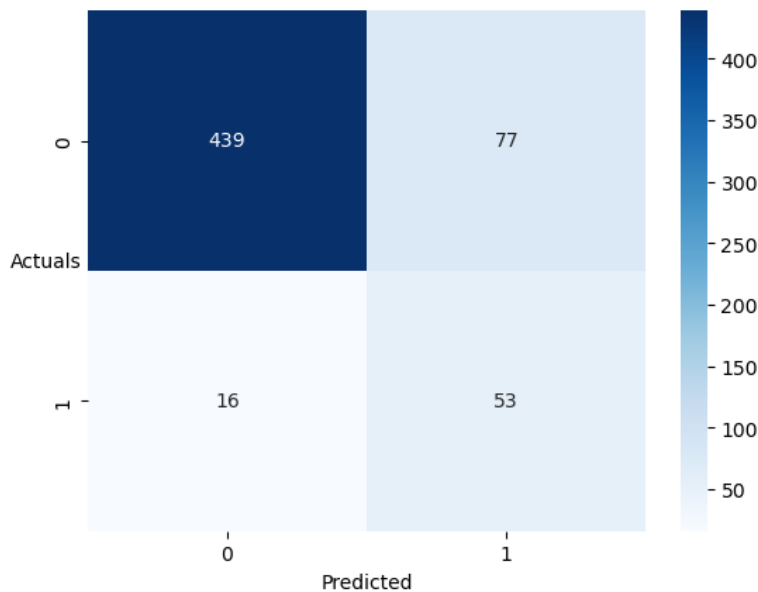
AUC Score: 0.8325835079502033



1.6 Validate the Model on Test Dataset and state the performance metrics. Also state interpretation from the model

The Optimized Logit Model has been validated on Test Dataset. The Confusion Matrix and Classification Report for the Test Dataset are as under:

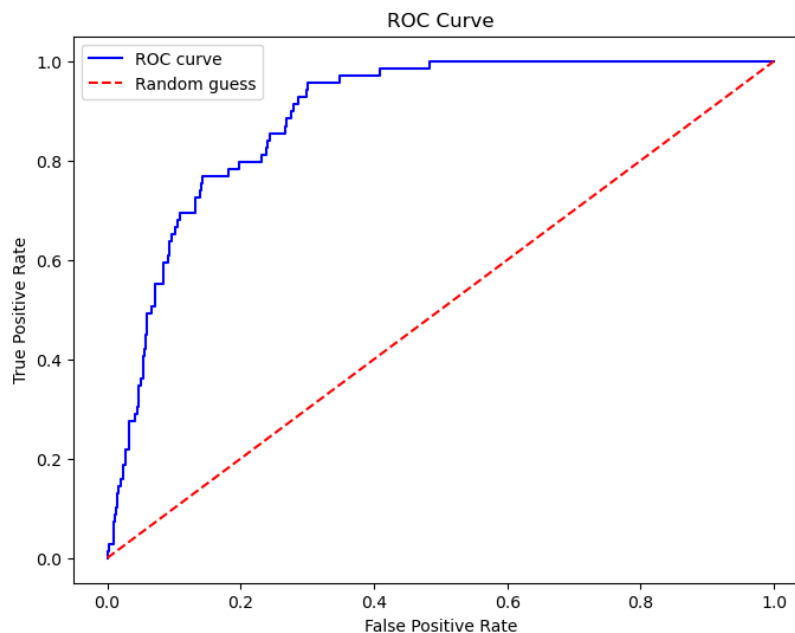
Confusion Matrix:



Classification Report:

	precision	recall	f1-score	support
0	0.965	0.851	0.904	516
1	0.408	0.768	0.533	69
accuracy			0.841	585
macro avg	0.686	0.809	0.718	585
weighted avg	0.899	0.841	0.860	585

The AUC Score of the Optimized Logit Model on test dataset is 0.89 and ROC Curve is as under:



Interpretations from the Model:

It is seen that the Model has a good Recall of 0.768 (on Default parameter) meaning that the data is effectively capturing 76.80% of the defaults explained by the Model. High Recall is crucial to Bank Defaults since missing a default can mean significant losses to the Bank.

Further, the Model has F1 Score of 0.841, which indicates that the Model is achieving both high Precision (low false positive rate) and high recall (low false negative rate). F1 score is particularly useful in case of imbalanced datasets where one probability (default) is much lower than the other (no default).

1.7 Build a Random Forest Model on Train Dataset. Also showcase your model building approach

Model Building Approach:

Random Forest Classifier was set with the following initial parameters:

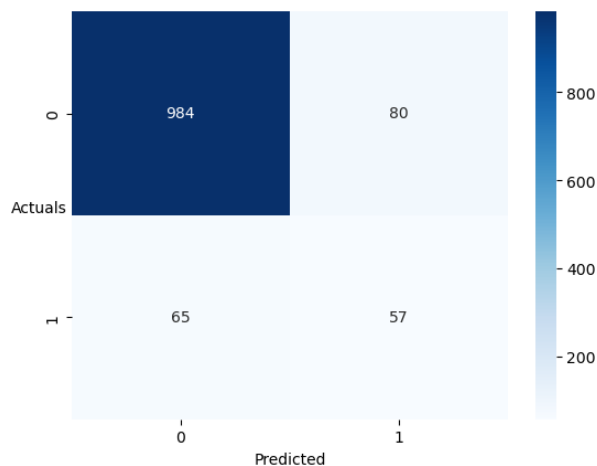
- i) max_depth: [3, 5, 7], maximum depth of the decision trees in the Random Forest to 7
- ii) min_samples_leaf: [5, 10, 15], Specifies the minimum number of samples required to be at a leaf node. This parameter can prevent the model from overfitting by setting a minimum size for leaf nodes.
- iii) min_samples_split: [15, 30, 45], Specifies the minimum number of samples required to split an internal node. This parameter can also help prevent overfitting by setting a minimum size for nodes to be split.
- iv) n_estimators: [25, 50], Sets the number of trees in the Random Forest. Increasing the number of trees can improve the performance of the model, but it also increases computation time.

Thereafter, the train and test data was fit into the Random Forest Model. Grid search was done to find the best parameters, which are as under:

```
{'max_depth': 7,  
'min_samples_leaf': 15,  
'min_samples_split': 15,  
'n_estimators': 25}
```

Thereafter predictions were done on Random Forest Model on Train Dataset. The Confusion Matrix and Classification Report are as under:

Confusion Matrix on Train Dataset with best Parameters:

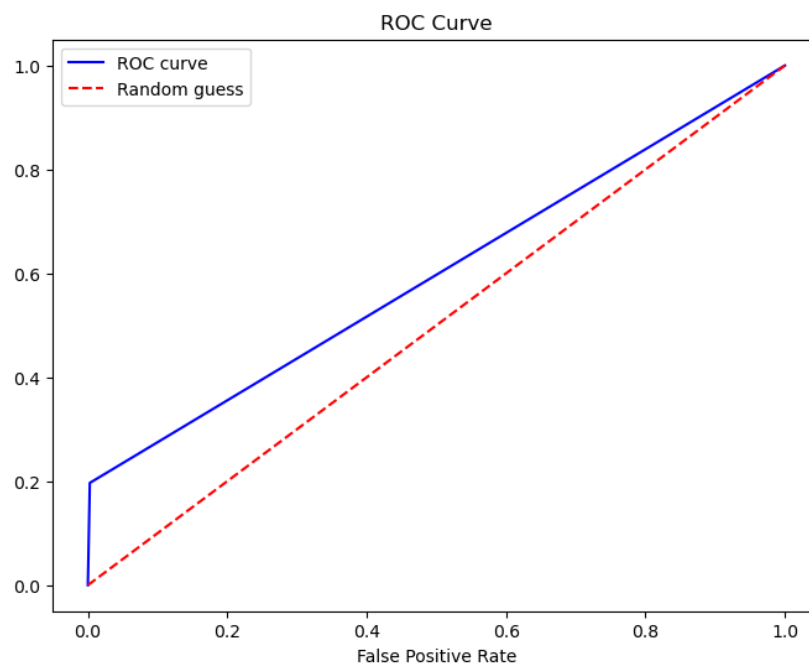


Classification Report on Train Dataset with best Parameters:

	precision	recall	f1-score	support
0	0.91	1.00	0.95	1064
1	1.00	0.16	0.28	122
accuracy			0.91	1186
macro avg	0.96	0.58	0.62	1186
weighted avg	0.92	0.91	0.89	1186

The AUC Score of the Random Forest Model on train dataset is 0.5969 and ROC Curve is as under:

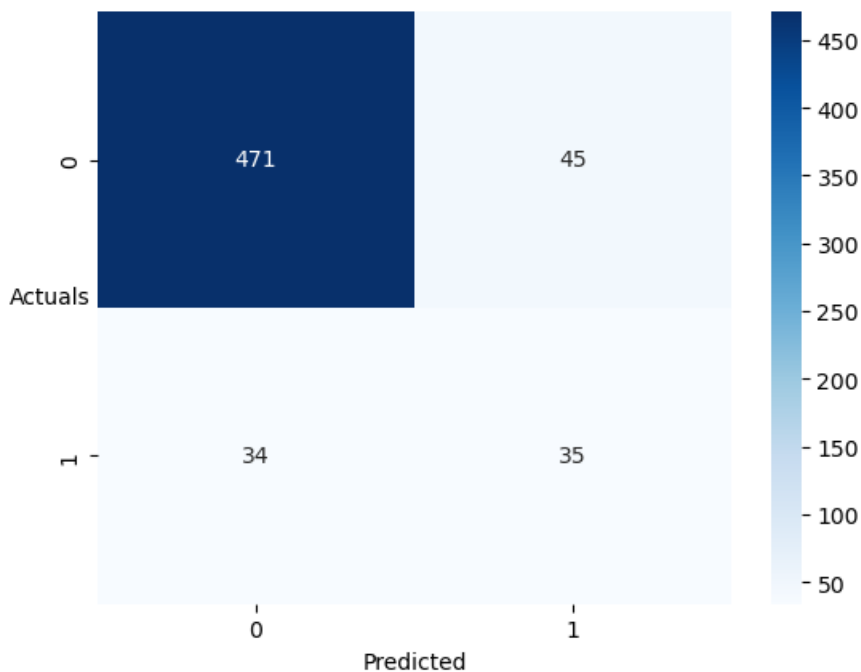
AUC Score: 0.5969508813016147



1.8 Validate the Random Forest Model on test Dataset and state the performance metrics. Also state interpretation from the model

The Optimized Random Forest was validated on Test Dataset. The Confusion Matrix and Classification Report are as under:

Confusion Matrix on Test Dataset with best Parameters:

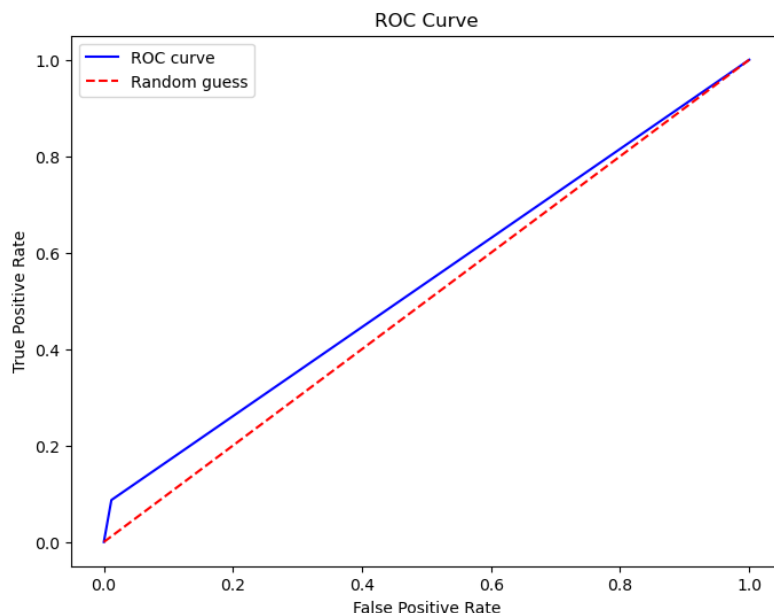


Classification Report on Test Dataset with best Parameters:

	precision	recall	f1-score	support
0	0.89	1.00	0.94	516
1	0.75	0.04	0.08	69
accuracy			0.89	585
macro avg	0.82	0.52	0.51	585
weighted avg	0.87	0.89	0.84	585

The AUC Score of the Random Forest Model on test dataset is 0.5377 and ROC Curve is as under :

AUC Score: 0.5376643073811931



Interpretations from the Model:

It is seen that the Model has a Recall of 0.04 (on Default parameter) meaning that the data is effectively capturing only 4% of the defaults explained by the Model. High Recall is crucial to Bank Defaults since missing a default can mean significant losses to the Bank.

Further, the Model has F1 Score of 0.08, which indicates that the Model is achieving both low Precision (low false positive rate) and low recall (low false negative rate). F1 score is particularly useful in case of imbalanced datasets where one probability (default) is much lower than the other (non default).

1.9 Build a Linear Discriminant Analysis (LDA) Model on Train Dataset. Also showcase your model building approach

Model Building Approach:

Firstly, LDA Model was fit into the data (X_train and y_train) and predictions were done on LDA Model on train and test data. The Classification Reports on train and test data are as under:

Classification Report on Train Dataset:

	precision	recall	f1-score	support
0	0.92	0.96	0.94	1064
1	0.47	0.28	0.35	122
accuracy			0.89	1186
macro avg	0.69	0.62	0.65	1186
weighted avg	0.87	0.89	0.88	1186

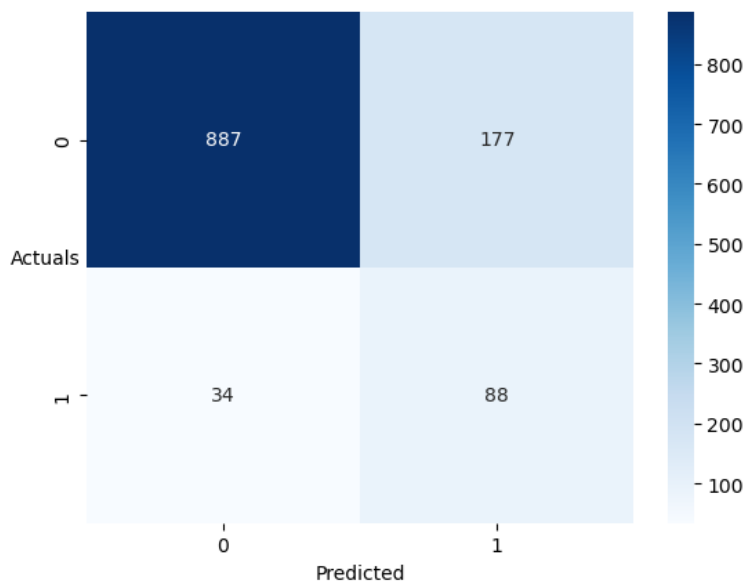
Classification Report on Test Dataset:

	precision	recall	f1-score	support
0	0.92	0.95	0.93	516
1	0.50	0.35	0.41	69
accuracy			0.88	585
macro avg	0.71	0.65	0.67	585
weighted avg	0.87	0.88	0.87	585

Next, the LDA Model was optimized for optimum threshold, which is found to be 0.12. LDA has been built on Train Dataset with the optimum threshold i.e. 0.12.

The Confusion Matrix and Classification Report with the optimum threshold are as under:

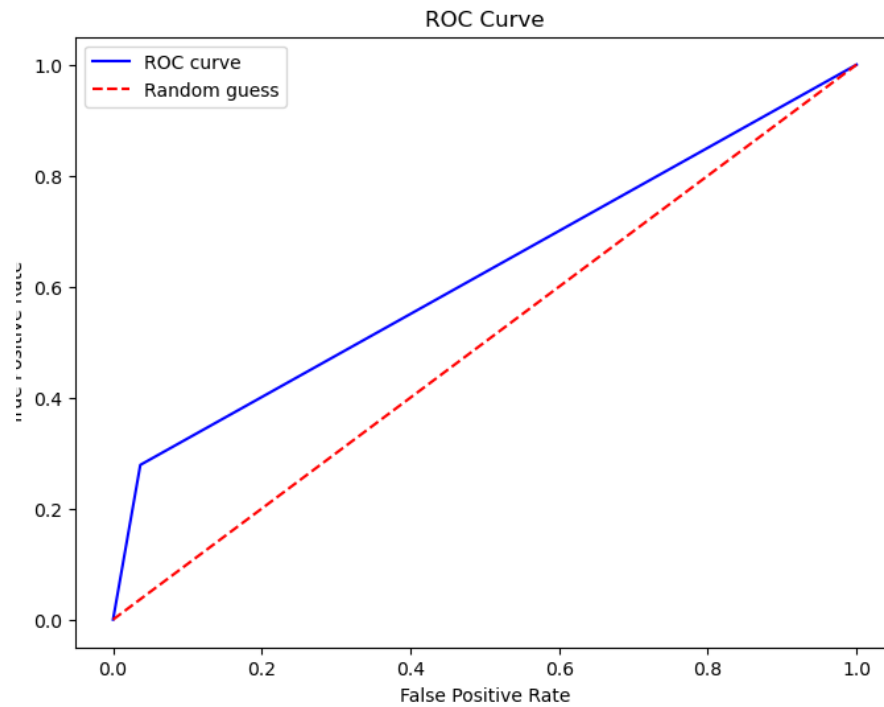
Confusion Matrix of the Optimized Train Dataset:



Classification Report of the Optimized Train Dataset:

	precision	recall	f1-score	support
0	0.963	0.834	0.894	1064
1	0.332	0.721	0.455	122
accuracy			0.822	1186
macro avg	0.648	0.777	0.674	1186
weighted avg	0.898	0.822	0.849	1186

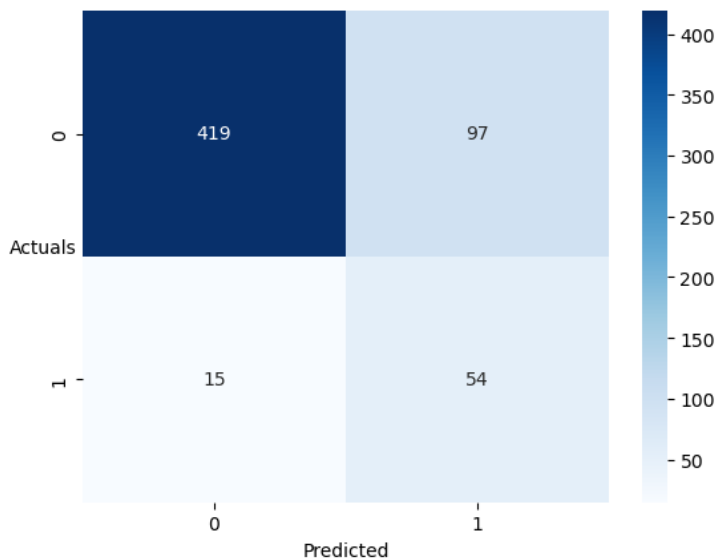
The AUC Score of the optimized LDA Model on train dataset is 0.6210 and ROC Curve is as under :



1.10 Validate the LDA Model on test Dataset and state the performance metrics. Also state interpretation from the model

Optimized LDA Model has been validated on Test Dataset. The Confusion Matrix and Classification Report for Test Dataset are as under:

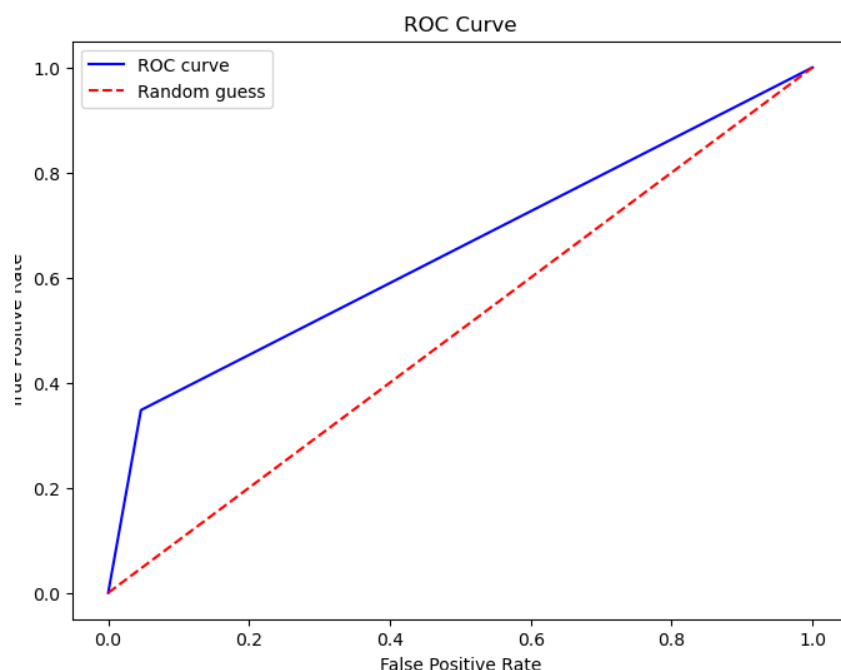
Confusion Matrix of the Optimized Test Dataset:



Classification Report of the Optimized Test Dataset:

	precision	recall	f1-score	support
0	0.965	0.812	0.882	516
1	0.358	0.783	0.491	69
accuracy			0.809	585
macro avg	0.662	0.797	0.687	585
weighted avg	0.894	0.809	0.836	585

The AUC Score of the optimized LDA Model on test dataset is 0.6506 and ROC Curve is as under:



Interpretations from the Model: Optimized LDA Model has a Recall of 0.783 (on Default parameter) meaning that the data is effectively capturing 78.3% of the defaults explained by the Model. High Recall is crucial to Bank Defaults since missing a default can mean significant losses to the Bank.

Further, the Model has F1 Score of 0.491 on Default , which indicates that the Model is achieving medium Precision (false positive rate) and medium recall (false negative rate). F1 score is particularly useful in case of imbalanced datasets where one probability (default) is much lower than the other (non default).

1.11 Compare the performances of Logistic Regression, Random Forest, and LDA models (include ROC curve)

Comparison of Models: Comparison of all the aforesaid Models is tabulated as under:

S.No.	Model	Accuracy (Train) (For Default)	Accuracy (Test) (For Default)	Recall (Test) (For Default)	F1 Score (Test) (For Default)	AUC Score of the Model (Test)
1.	Logistic Regression (Optimized)	0.852	0.841	0.768	0.533	0.89
2.	Linear Discriminant Analysis (Optimized)	0.822	0.809	0.783	0.491	0.65
3.	Random Forest Classifier (With best parameters)	0.91	0.89	0.04	0.08	0.54

Inference:

Among all the Models, Logistic Regression Model (Optimized) is the best model with Recall of 0.768 (on Default parameter) meaning that the Model is effectively capturing 76.80% of the defaults. High Recall is crucial to Bank Defaults since missing a default can mean significant losses to the Bank.

Further, the Model has F1 Score of 0.841, which indicates that the Model is achieving both high Precision (low false positive rate) and high recall (low false negative rate). F1 score is particularly useful in case of imbalanced datasets where one probability (default) is much lower than the other (no default).

The Model has a good AUC Score of 0.89 on test data. It is also observed that the Model is not overfitting because Accuracy on train data is 0.852 and on test data is 0.841.

LDA Model has a lower Accuracy, Recall and a lower AUC Score of 0.65. Further, Random Forest Model has a very low Recall (0.04), very low F1 Score (0.08) and a lower AUC Score of 0.54.

Therefore, it is observed that Optimized Logistic Regression Model is the best Model.

1.12 Conclusions and Recommendations:

- i) The crux of business problem is to predict the probability of default of companies based on our selected model so as to avoid the customers with the high probability of default based on our selected Model. Optimized Logistic Regression Model being our best selected model, predictions have been made with our selected variables with the coefficients as under:

```
Coefficients: Intercept                -3.343387e+00
Total_debt_to_Total_net_worth      1.353483e+02
Fixed_Assets_Turnover_Frequency    1.517181e+01
Tax_rate_A                        -5.779995e+00
RD_expense_rate                    1.730003e-10
Cash_Turnover_Rate                 -1.513670e-10
dtype: float64
```

It may be seen that variable 'Total Debt to Total Net Worth' is the most important variable impacting the probability of default, followed by 'Tax Rate A'.

- ii) Based on the Optimized Logistic Regression, the probability of default was calculated for the companies given in the data.

Top 10 companies and their probability of default is as under:

1073	0.726774
1873	0.700275
926	0.692208
238	0.691236
449	0.689500
605	0.684970
704	0.674362
466	0.657530
1644	0.637736
563	0.608078

10 Companies having least probability of default are as under:

1179	0.003407
72	0.002903
645	0.002890
354	0.002813
1879	0.002371
880	0.002137
351	0.001905
156	0.001896
571	0.001889
442	0.000803

dtype: float64

- iii) The Bank is recommended to maintain a risk profile of customers based on the predictions of Probability of Default based on best model. The Bank may avoid lending

to customer with high probability of Default and focusing at the same on customers having least probability of default.

